# Second order ODE's

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**Homogeneous :** $P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0$ ← 0 if homogeneous

If $y_1(x)$ and $y_2(x)$ are both sols,

$y(x) = c_1y_1(x) + c_2y_2(x)$ is also a sol ($c_1$, $c_2$ constants)

general sol : $y(x) = c_1y_1(x) + c_2y_2(x)$

↑ linearly independent sols

$ay'' + by' + c = 0$ ⇒ auxiliary eqn : $ar^2 + br + c = 0$

Solve auxiliary eqn : $r_1, r_2$

$y = e^{rx}$

$y' = re^{rx}$

$y'' = r^2e^{rx}$

case 1 : $b^2 - 4ac > 0$ ($r_1, r_2$ distinct & real)

general sol : $y = c_1e^{r_1x} + c_2e^{r_2x}$

case 2 : $b^2 - 4ac = 0$ ($r_1 = r_2$)

general sol : $y = c_1e^{rx} + c_2xe^{rx}$

case 3 : $b^2 - 4ac < 0$ ($r_1, r_2$ distinct & imaginary)

$r_1 = \alpha + i\beta$

$r_2 = \alpha - i\beta$

general sol : $y = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x)$ (real part, imaginary part)

**Nonhomogeneous :** $ay'' + by' + cy = G(x)$ (not 0)

complementary eqn : $ay'' + by' + cy = 0$

general sol : $y(x) = y_p(x) + y_c(x)$ (particular sol, general sol)

## Method of undetermined coefficients

$ay'' + by' + cy = G(x)$

guess : $y_p(x)$ is a polynomial of same degree as G

sub into eqn. and determine coefficients

polynomial → $Ax^n + Bx^{n-1} + \dots$

$\cos(mx)$ / $\sin(mx)$ → $A\cos(mx) + B\cos(mx)$

exponential ($e^{kx}$) → $Ae^{kx}$

product → take product

e.g. $e^{kx}P(x)\cos(mx) \rightarrow e^{kx}Q(x)\cos(mx) + e^{kx}R(x)\sin(mx)$

sum → principle of superposition

$y = y_c + y_{p_1} + y_{p_2}$, $y_{p_1}$, $y_{p_2}$ are particular sols to $G_1(x)$, $G_2(x)$

If $y_p(x)$ is a soln of complementary eqn ($y_c(x)$), then multiply by $x$ (or $x^2$, $x^3$, ...)

e.g. $y'' + y' - 2y = x^2$

aux eqn : $r^2 + r - 2 = 0$

$(r-1)(r+2) = 0$

$r = 1, -2$

$y_c = c_1e^x + c_2e^{-2x}$

particular soln : $y_p(x) = Ax^2 + Bx + C$

$(2A) + (2Ax + B) - 2(Ax^2 + Bx + C) = x^2$

$-2Ax^2 + (2A - 2B)x + (2A + B - 2C) = x^2$

equate coefficients :

$-2A = 1$

$2A - 2B = 0$

$2A + B - 2C = 0$

$\therefore A = -\frac{1}{2}, B = -\frac{1}{2}, C = -\frac{3}{4}$

$y_p(x) = -\frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}$

$y = y_c(x) + y_p(x)$

$= c_1e^x + c_2e^{-2x} - \frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}$

## Method of variation of parameters

$ay'' + by' + c = G$

solved complementary eqn : $y_c(x) = c_1y_1(x) + c_2y_2(x)$

look for particular soln by replacing parameters $c_1, c_2$ with functions $u_1(x), u_2(x)$

$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$ (*)

$y_p' = (u_1'y_1 + u_2'y_2) + (u_1y_1' + u_2y_2')$

Impose condition : $u_1'y_1 + u_2'y_2 = 0$ (*)

$y_p'' = u_1'y_1' + u_2'y_2' + u_1y_1'' + u_2y_2''$

sub into diff. eqn :

$a(u_1'y_1' + u_2'y_2' + u_1y_1'' + u_2y_2'') + b(u_1y_1' + u_2y_2') + c(u_1y_1 + u_2y_2) = G$

$u_1(ay_1'' + by_1' + cy_1) + u_2(ay_2'' + by_2' + cy_2) + a(u_1'y_1' + u_2'y_2') = G$

$ay_1'' + by_1' + cy_1 = 0$, $ay_2'' + by_2' + cy_2 = 0$

$y_1, y_2$ are sols of complementary eqn

$\therefore a(u_1'y_1' + u_2'y_2') = G$ (**)

(*) and (**) form system of two equations in $u_1'$ and $u_2'$

Integrate to get $u_1$, $u_2$, and particular solution (*)

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e.g. $y'' + y = \tan x$, $0 < x < \frac{\pi}{2}$

aux eqn : $r^2 + 1 = 0 \Rightarrow r = \pm i$

$y(x) = c_1\sin x + c_2\cos x$

$y_p(x) = u_1(x)\sin x + u_2(x)\cos x$

$y_p' = \underbrace{(u_1'\sin x + u_2'\cos x)}_{\text{set this to 0}} + (u_1\cos x - u_2\sin x)$

$y_p'' = (u_1'\cos x - u_2'\sin x) + (-u_1\sin x - u_2\cos x)$

for $y_p$ to be a sol, $y_p'' + y_p = \tan x$

$u_1'\cos x - u_2'\sin x = \tan x$

$\begin{cases} u_1'\sin x + u_2'\cos x = 0 \rightarrow u_1'\sin^2 x + u_2'\sin x\cos x = 0 \\ u_1'\cos x - u_2'\sin x = \tan x \rightarrow u_1'\cos^2 x - u_2'\sin x\cos x = \cos x\tan x \end{cases}$

$u_1'(\sin^2 x + \cos^2 x) = \cos x\tan x$

$u_1' = \cos x\tan x = \sin x \qquad u_1 = -\cos x$

$u_2' = \frac{-u_1'\sin x}{\cos x}$

$= -\frac{\sin^2 x}{\cos x}$

$= -\frac{1 - \cos^2 x}{\cos x}$

$= -\sec x + \cos x \qquad u_2 = \sin x - \ln(\sec x + \tan x)$

$y_p(x) = -\cos x\sin x + [\sin x - \ln(\sec x + \tan x)]\cos x$

$= -\cos x\ln(\sec x + \tan x)$

$y(x) = c_1\sin x + c_2\cos x - \cos x\ln(\sec x + \tan x)$